Syllabus Details

Syllabus ID:	12580
Syllabus Name:	**Front-End web development with React_Phát triển web Front-End với React**
Course Name English:	**Front-End web development with React**
Subject Code:	**FER202**
Learning-Teaching Method:	Project based learning, Problem based learning, Active learning, In-class lecture
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	Study hour (150h) = 45h contact hours + 1h final exam + 85' practical exam + 104h self-study
Pre-Requisite:	WED201c
Description:	Learn front-end web development for implementing a multi-platform solution: - Overview of client-side web UI frameworks, focusing on Bootstrap. Students will explore grids, responsive design, Bootstrap CSS, and JavaScript components. - Basics of Node.js, npm - JavaScript-based front-end development with the React library: ES6 for React applications, Reactstrap for Bootstrap-based responsive UI, React components, React Router for single-page applications, and designing controlled forms. - Hooks to leverage React features like state and lifecycle in functional components, reducing reliance on class components. - Flux architecture and Redux, covering Redux principles and building React-Redux powered applications. Fetch for client-server communication, integrating REST APIs on the server side. - Modern syntax and patterns for ReactJS development, paired with foundational knowledge to ensure long-term relevance. - AI Integration: Use of AI tools (e.g., code generation with GenAI) to enhance development efficiency, with emphasis on transparency and ethical usage. - Assessment Methods: Evaluation through practical labs, assignments, and reviews using stratified sampling, scored via Likert scales (1-5) for syntax, logic, customization, and transparency, converted to a 10-point scale.
StudentTasks:	- Students must attend at least 80% of contact slots to be eligible for the final examination. - Students are responsible for completing all exercises assigned by the instructor in class or at home and submitting them on time. - Regularly check announcements on FU FLM at https://flm.fpt.edu.vn/ for the latest course updates, including assignment submission deadlines and feedback on assignments and project work. - AI-Related Task: Utilize AI tools (e.g., GenAI for code generation or debugging) responsibly in assignments and projects, and clearly document their usage (e.g., prompts, outputs) in submissions to ensure transparency and ethical application.
Tools:	- Visual Studio Code (https://code.visualstudio.com) - Internet - Git (Bitbucket/Github source management) - Node.js AI tools - Gemini (https://gemini.google.com/app) - Grok (https://grok.com/) - ChatGPT (https://chatgpt.com/)
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	359/QĐ-ĐHFPT dated 04/09/2025
IsApproved:	**True**
Note:	
Is Scored:	**True**

MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	4/9/2025

3 material(s)

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
1	https://getbootstrap.com/docs/5.3/getting-started/introduction/						True	False	True	
2	https://react.dev/learn						True	False	True	
3	https://react.dev/learn/start-a-new-react-project#nextjs						False	False	True	

9 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Describe Front-end Web UI Frameworks overview
2	CLO2	Install React application and Understand React component
3	CLO3	Explain JSX and ES6
4	CLO4	Explain the Bootstrap, React-boostrap
5	CLO5	Discuss the React Component Life Cycle, Forms and Routes
6	CLO6	Discuss Hook, Event Handling and crafting reusable component in React
7	CLO7	Discuss fetch and catching data in Client-Server Communication and Code Splitting Using Lazy Components and Suspense
8	CLO8	Discuss Flow Architecture and Redux
9	CLO9	Apply AI tools in React development and evaluate their ethical use and transparency

View mapping of CLOs to PLOs

Download All Student Material

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
1	Assignment Introduction with AI Integration	Offline	CLO1-CLO9	I, T	Assignment brief	FER202	Study assignment brief, brainstorm app idea with AI, note CLO1-CLO9 requirements	
2	Course Introduction 1.1 Full Stack Web Development: The Big Picture 1.2 React 19 Introduction	Offline	CLO1	I	Slides		Study slides, note key concepts of full-stack and React 19	
3	1.3 Git - Setting up Git - Basic Git Commands - Setting up an Online Git repository 1.4 Node.js and npm - Setting up Node.js and npm - Basics of Node.js and npm	Offline	CLO1	I, T, U	Slides		Install Git, practice basic commands (init, add, commit), study slides Create GitHub account, push repo, install Node.js/npm, study slides	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
4	Rendering with JSX + ES 6 2.1 ES6	Offline	CLO3	I, T	Slides		Study ES6 features (arrow functions, destructuring), practice examples	
5	2.2 ES6 (cont'd) - Using AI to generate ES6 code snippets	Offline	CLO3, CLO9	T, U	Slides, exercises, AI tool		Use GenAI (e.g., ChatGPT) to generate ES6 snippets, compare with slides, do exercises	
6	2.3 ES6 (cont'd)	Offline	CLO3	U	Slides, exercises		Complete ES6 exercises, apply to simple React components	
7	3.1 Rendering HTML 3.2 Creating your own JSX elements	Offline	CLO3	T, U	Slides		Study JSX syntax, create basic JSX elements, practice rendering	
8	3.3 Using JavaScript expressions	Offline	CLO3	U	Slides		Use JS expressions in JSX, practice with examples	
9	3.4 Building fragments of JSX LAB 1: Install React + JSX Basics	Offline	CLO1, CLO2, CLO3	T, U	Slides, lab questions		Install React app with CRA, build JSX fragments, complete Lab 1	
10	Review Lab 1	Offline	CLO1, CLO2, CLO3, CLO9	U	Lab submissions		Submit Lab 1, use AI to debug code, prepare questions for review	
11	Review Lab 1 (td)	Offline	CLO1, CLO2, CLO3, CLO9	U	Lab submissions		Submit Lab 1, use AI to debug code, prepare questions for review	
12	Bootstrap and React-bootstrap 4.1 Bootstrap & React-Bootstrap Introduction - Set up a project with Bootstrap & React-Bootstrap - Configure a web project to use Bootstrap and React-Bootstrap	Offline	CLO4	I, T	Slides		Study Bootstrap/React-Bootstrap, set up project with both	
13	4.2 Bootstrap Grid System	Offline	CLO4	T, U	Slides		Practice grid system layouts, apply to sample UI	
14	4.3 Bootstrap and React-Bootstrap CSS Components - Navigation and Navigation Bar - Icon and Fonts - User Input: Buttons and Forms - Tabular data (Table)	Offline	CLO4	T, U	Slides, exercises		Build UI with BS components (nav, buttons), do exercises	
15	5.1 Bootstrap JavaScript Components - Tabs, Pills and Tabbed Navigation	Offline	CLO4	T, U	Slides, exercises		Implement JS components (tabs/pills), test functionality	
16	5.2 Hide and Seek: Collapse and Accordion	Offline	CLO4	U	Slides, exercises		Add collapse/accordion to UI, practice interactivity	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
17	5.3 Compare betweent Bootstrap and React-Bootstrap	Offline	CLO4	T, U	Slides, exercises		Build UI with BS/React-BS	
18	LAB 2: Bootstrap and React-Bootstrap UI	Offline	CLO4	T, U	Slides, lab questions		Compare in Lab 2, submit work	
19	Review Lab 2	Offline	CLO4, CLO9	U	Lab submissions		Submit Lab 2, use AI to optimize UI code, prepare for review	
20	Review Lab 2 (td)	Offline	CLO4, CLO9	U	Lab submissions		Submit Lab 2, use AI to optimize UI code, prepare for review	
21	React Component 6.1 What's react component? How to build it? 6.2 Component Properties	Offline	CLO2	I, T, U	Slides		Study component basics, build a simple component with props	
22	6.3 Component State - Setting initial component state - Creating component state - Merging component state	Offline	CLO2	I, T	Slides, exercises		Practice setting initial state in components, do exercises Build components with state (Class/Functional), merge state, practice	
23	Event Handling 7.1 Declaring event handlers - Binding handlers to elements - Using synthetic event objects - Understanding event pooling	Offline	CLO6	I, T, U	Slides, exercises		Study event handling, bind handlers to elements, practice Practice synthetic events, explore pooling, do exercises	
24	7.2 Using event handler context and parameters - Declaring inline event handlers LAB 3: Class vs Functional Components	Offline	CLO2, CLO6	T, U	Slides, exercises, lab questions		Add inline handlers, compare Class/Functional in Lab 3, submit work	
25	Review Lab 3	Offline	CLO2, CLO6, CLO9	U	Lab submissions		Submit Lab 3, use AI to refactor code, prepare for review	
26	Review Lab 3 (td)	Offline	CLO2, CLO6, CLO9	U	Lab submissions		Submit Lab 3, use AI to refactor code, prepare for review	
27	8.1 Passing property values 8.2 Stateless Components	Offline	CLO5	T, U	Slides		Build stateless components, pass props, practice	
28	Container components 8.3 Providing and consuming contex	Offline	CLO5	T, U	Slides		Implement context in components, test data sharing, practice	
29	8.4 Render props 8.5 Refactoring class components using hooks	Offline	CLO6	T, U	Slides		Refactor class components with Hooks, practice render props	
30	Getting Started with Hooks 9.1 Maintaining state using Hooks	Offline	CLO6	I, T	Slides		Study useState, implement in a component, practice	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
31	9.2 Performing initialization and cleanup actions	Offline	CLO6	T, U	Slides, exercises		Use useEffect for lifecycle tasks, do exercises	
32	9.3 Sharing Data with Context Hooks	Offline	CLO5, CLO6	T, U	Slides		Implement context Hooks, share data across components, practice	
33	9.4 Using Reducer Hooks for State Management LAB 4: Hooks & Events	Offline	CLO6	T, U	Slides, lab questions		Use useReducer for complex state, complete Lab 4 with events	
34	Review Lab 4	Offline	CLO6, CLO9	U	Lab submissions		Submit Lab 4, use AI to enhance event logic, prepare for review	
35	Review Lab 4 (td)	Offline	CLO6, CLO9	U	Lab submissions		Submit Lab 4, use AI to enhance event logic, prepare for review	
36	Progress Test - 01	Offline	CLO1-CLO6	T, U	Test questions		Study materials, complete Progress Test 1	
37	Handling Navigation with Routes 10.1 Declaring routes - Decoupling route declaration	Offline	CLO5	I, T	Slides		Study React Router, set up basic routes, practice	
38	10.2 Handling route parameters - Resource IDs in routes - Optional parameters	Offline	CLO5	T, U	Slides		Add route parameters, test navigation, practice	
39	10.3 Using link components - Basic linking - URL and query parameters	Offline	CLO5, CLO7	T, U	Slides		Build routed app with and lazy loading, complete Lab 5	
40	Code Splitting Using Lazy Components and Suspense 11.1 Using Lazy API 11.2 Using the Suspense component -Simulating latency	Offline	CLO7	T, U	Slides		Use React.lazy and Suspense, simulate latency, do exercises	
41	11.3 Avoiding lazy components 11.4 Exploring lazy pages and routes	Offline	CLO7	T, U	Slides, exercises		Apply lazy loading to routes, test performance	
42	Client - Server Communication 12.1 Set up a simple server that makes data available for clients LAB 5: Routes & Lazy Loading	Offline	CLO7	I, T	Slides, lab questions		Set up JSON server, study client-server basics, complete Lab 5	
43	Review Lab 5	Offline	CLO5, CLO7, CLO9	U	Lab submissions		Submit Lab 5, use AI to optimize routing, prepare for review	
44	Review Lab 5 (td)	Offline	CLO5, CLO7, CLO9	U	Lab submissions		Submit Lab 5, use AI to optimize routing, prepare for review	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
45	12.2 Access the data from the server using a browser 12.3 Use the json-server as a simple static web server	Offline	CLO7	T, U	Slides		Fetch data from server, display in app, practice	
46	Fetching and Catching Data with React 13.1 When to fetch and store data in React. How to use window.fetch.	Offline	CLO7	I, T	Slides		Study fetch, implement in app, practice	
47	13.2 What promises are. How async/await works.	Offline	CLO7	T, U	Slides, exercises		Use async/await with fetch, practice exercises	
48	13.3 Getting Data with Axios How to simplify network requests with Axios. LAB 6: Fetching Data	Offline	CLO7	T, U	Slides, lab questions		Replace fetch with Axios, complete Lab 6	
49	Review Lab 6	Offline	CLO7, CLO9	U	Lab submissions		Submit Lab 6, use AI to refine fetch logic, prepare for review	
50	Review Lab 6 (td)	Offline	CLO7, CLO9	U	Lab submissions		Submit Lab 6, use AI to refine fetch logic, prepare for review	
51	Flow Architecture and Intrduction to Redux 14.1 Explain the Redux approach to implementing a variant of the Flow architecture.	Offline	CLO8	I, T	Slides		Study Redux principles, set up basic store, practice	
52	14.2 Install and Configure Redux.	Offline	CLO8	I, T	Slides		Install Redux, configure store, practice	
53	14.3 Define Redux Toolkit	Offline	CLO8, CLO9	T, U	Slides, AI tool		Use GenAI to generate Redux Toolkit code, integrate into app, practice	
54	Redux architecture 15.1 Split the reducer function into multiple simpler functions and combine the reducer functions 15.2 Use Redux Thunk middleware to return a function instead of an action	Offline	CLO8	T, U	Slides		Split reducers, implement Redux Thunk for async, practice	
55	15.3 Use a logger middleware to print a log of actions initiated on the Redux store	Offline	CLO8	T, U	Slides		Add logger middleware, monitor actions, practice	
56	Progress Test - 02	Offline	CLO7-CLO8 CLO1-CLO8	T, U	Test questions		Study materials, complete Progress Test 2	
57	Assignment Evaluate	Offline	CLO1-CLO9	U	Assignment submissions		Submit assignment, prepare for review with AI-optimized code	
58	Assignment Evaluate	Offline	CLO1-CLO9	U	Assignment submissions		Submit assignment, prepare for review with AI-optimized code	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
59	Assignment Evaluate	Offline	CLO1-CLO9	U	Assignment submissions		Submit assignment, prepare for review with AI-optimized code	
60	Assignment Evaluate	Offline	CLO1-CLO9	U	Assignment submissions		Submit assignment, prepare for review with AI-optimized code	

0 Constructive question(s)

5 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
1	Assignment	on-going	1	15.0%	>0	Guided in tutorials (Sessions 1), cont. at home	CLO1-CLO9	AI prompt, Likert review		Full-stack app using React, AI tools, covering CLO1-CLO9	Individual or team work, guided by instructor, submission by deadline	Introduced in Session 1, Evaluate in Session 57-60.
2	Labs	on-going	6	20.0%	>0	Guided and reviewed in tutorial sesions; Cont. at home	Lab 1: CLO1, CLO2, CLO3 Lab 2: CLO4 Lab 3: CLO2, CLO6 Lab 4: CLO6, CLO9 Lab 5: CLO5, CLO7 Lab 6: CLO7, CLO9	Practical exercises using AI prompt, Likert review		Lab 1: React setup, JSX Lab 2: Bootstrap UI Lab 3: Components, Events Lab 4: Hooks Lab 5: Routes, Lazy Loading Lab 6: Fetching data	Teachers support tutorials, students complete independently, 1-1 review for 10 students per Lab	Reviewed in Sessions 10, 19, 25, 34, 43, 49 using AI assistance (CLO9). Submission via EduNext.
3	Practical Exam	on-going	1	20.0%	4	85'	CLO1-CLO9	Preferable to be marked by scripts		Build React app with AI integration, covering CLO1-CLO9	Supervised by proctor(s) sent by exam board	
4	Progress test	on-going	2	15.0%	>0	30' each	PT 1 CLO1 - CLO6 PT 2 CLO7 - CLO8	Multiple choice or essay	20-25 (MCQ) or 5-8 (essay) per PT	PT 1: Fundamentals, Components, Hooks PT 2: Client-Server, Redux	Supervised by instructor, online or in-class	PT 1 in Session 36, PT 2 in Session 56, reviewed post-grading by instructor.

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
5	Final exam	Final exam	1	30.0%	4	60'	CLO1-CLO9	multiple choice	50	all topics; more than 70% new questions (for the current semester);	supervised by proctor(s) sent by exam board	